Walkability and Typical Home Values Across U.S. ZIP Codes

Research question: In places where daily errands are easier to do on foot, are typical home values higher?
Data: EPA National Walkability Index (ZIP) merged with Zillow-style typical home values (`zillow_clean.csv`).

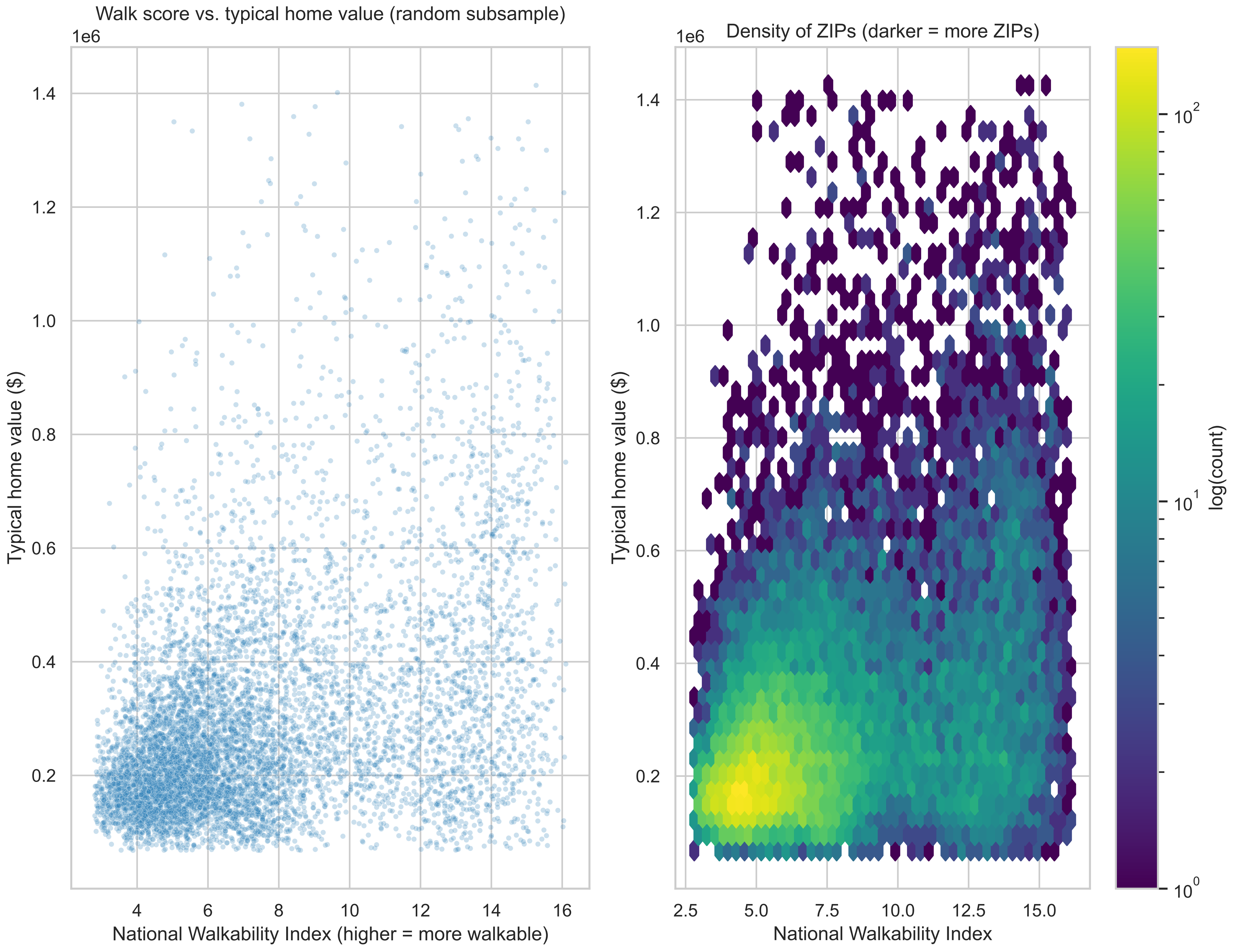
Cleaning highlights:
- ZIP codes standardized to 5 digits for a reliable merge.
- Rows with missing walk score, price, or state removed (0 rows).
- Extreme home values trimmed at the 1st–99th percentile to limit outlier leverage while keeping most of the market (68, 073–1,425,633).
- Walk scores trimmed at the 1st–99th percentile for robustness.

Final analytic sample: 18,535 ZIP codes after cleaning.

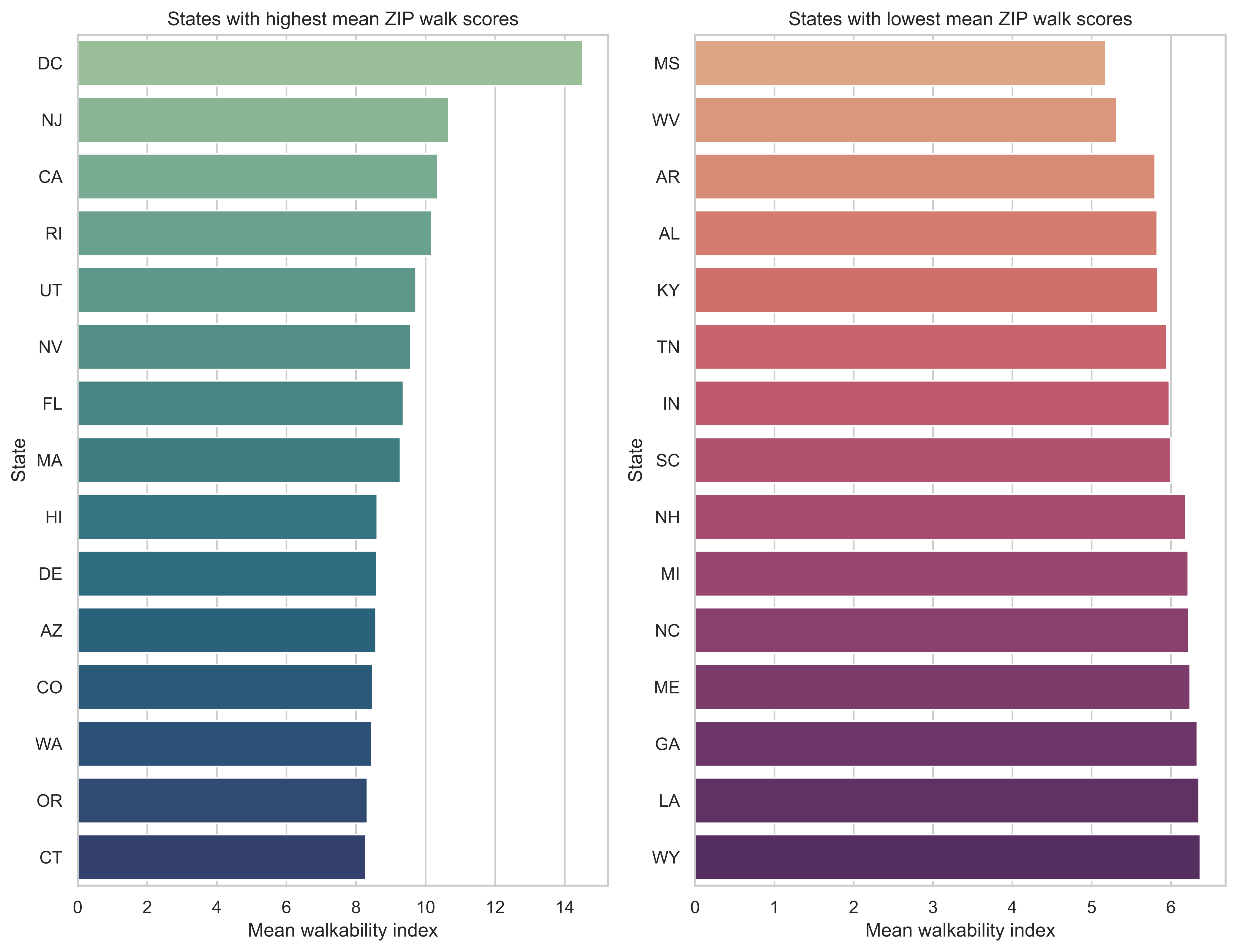
Why it matters: Walkability is a proxy for land-use intensity, amenities, and transport options. If prices rise with walkability, that signals how households trade off housing costs against neighborhood accessibility—useful for planners, buyers, and researchers studying urban form and affordability.

Summary statistics (ZIP-level, cleaned sample)

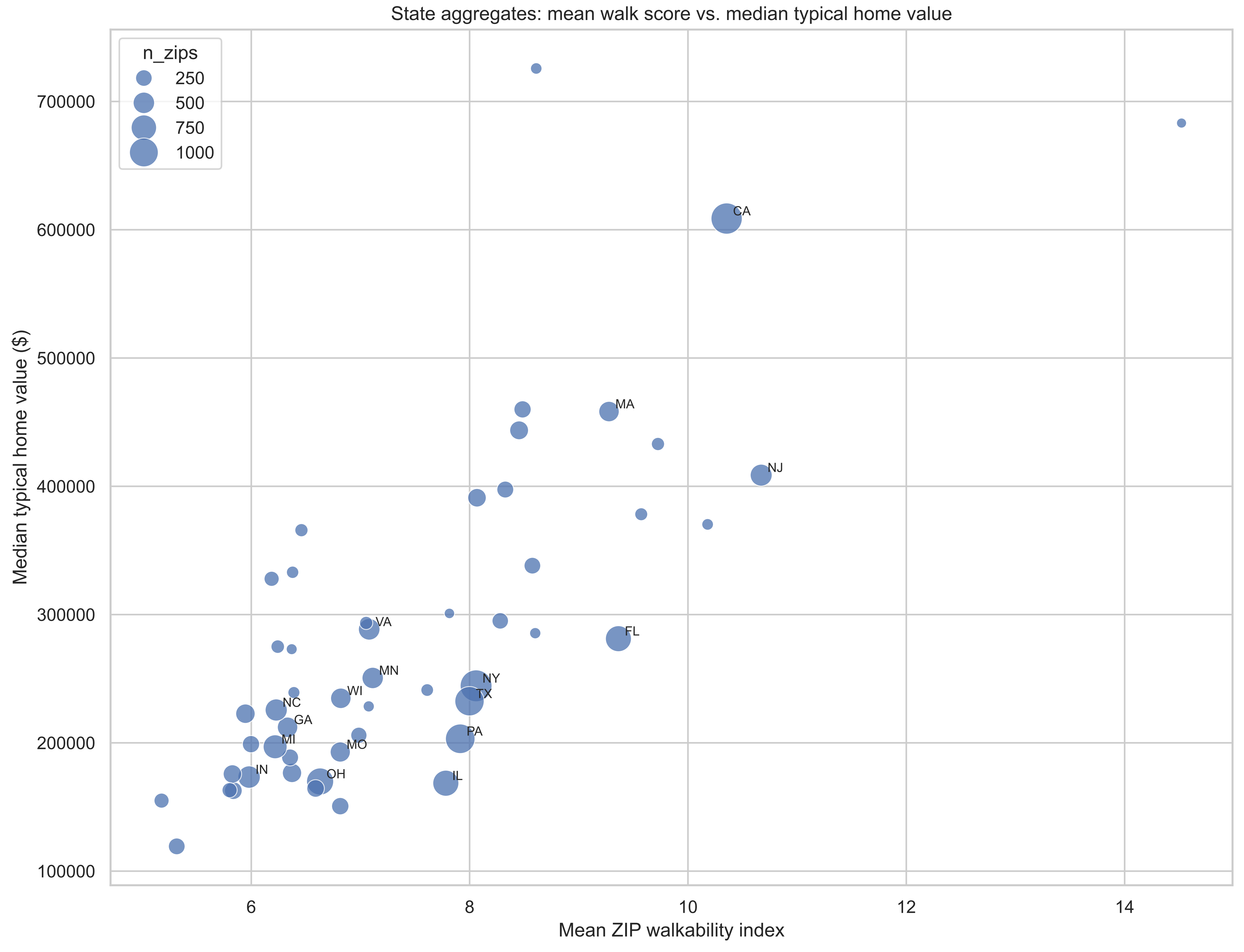
	NatWalkInd	home_value
count	18535.0	18535.0
mean	7.58	303321.87
std	3.38	203695.82
min	2.81	68073.46
5%	3.61	108063.44
25%	4.93	168508.18
50%	6.55	240619.27
75%	9.64	367222.56
95%	14.26	720676.56
max	16.12	1425633.21



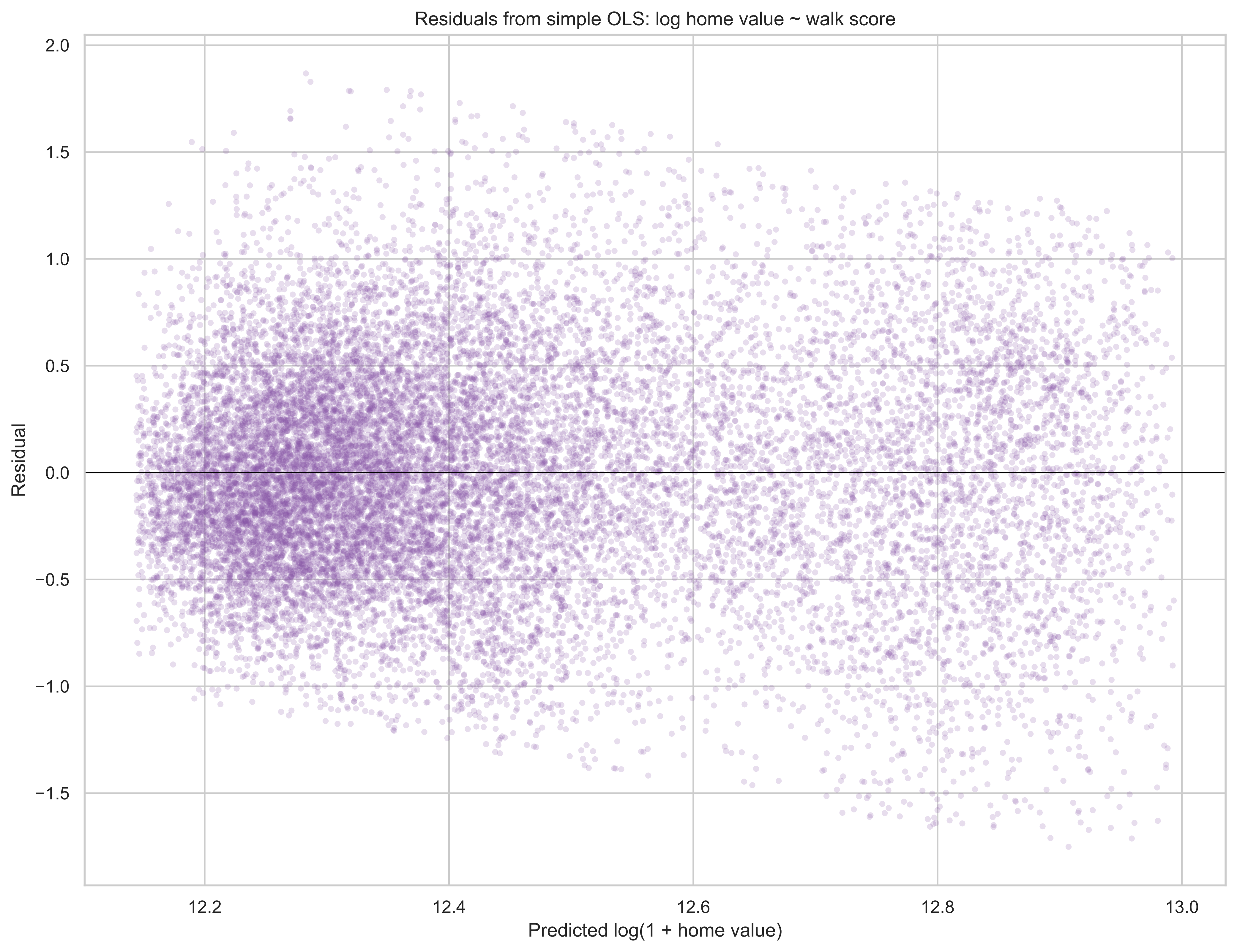
Walk score vs. typical home value (random subsample)
1e6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
Typical home value ($)
4
6
8
10
12
14
16
National Walkability Index (higher = more walkable)
Density of ZIPs (darker = more ZIPs)
1e6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
Typical home value ($)
2.5
5.0
7.5
10.0
12.5
15.0
National Walkability Index
10^2
10^1
10^0
log(count)



States with highest mean ZIP walk scores
DC
NJ
CA
RI
UT
NV
FL
MA
HI
DE
AZ
CO
WA
OR
CT
State
0
2
4
6
8
10
12
14
Mean walkability index
States with lowest mean ZIP walk scores
MS
WV
AR
AL
KY
TN
IN
SC
NH
MI
NC
ME
GA
LA
WY
State
0
1
2
3
4
5
6
Mean walkability index



State aggregates: mean walk score vs. median typical home value
n_zips
250
500
750
1000
CA
MA
NJ
VA
FL
MN
NY
WI
TX
NC
GA
PA
MI
MO
IN
OH
IL
700000
600000
500000
400000
300000
200000
100000
Median typical home value ($)
6
8
10
12
14
Mean ZIP walkability index


Residuals from simple OLS: log home value ~ walk score
Residual
Predicted log(1 + home value)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
12.2
12.4
12.6
12.8
13.0

```
Statistical analysis (ZIP-level, cleaned sample)

Pearson correlation (walk, price): r = 0.394, p = 0.00e+00
Spearman rank correlation: rho = 0.372, p = 0.00e+00

Home values are right-skewed; log transform stabilizes variance.
Pearson correlation (walk, log1p(price)): r = 0.378, p = 0.00e+00

Simple OLS: log1p(home_value) = beta0 + beta1 * NatWalkInd
  beta0 (intercept): 11.9653
  beta1 (per walk unit): 0.06377
  R-squared on log scale: 0.143

Interpretation: A positive association means ZIPs with higher walk scores tend to have higher typical prices, but correlation is not causation—dense, high-amenity places can be both walkable and expensive for many reasons.
```